

The Euclidian Algorithm

Chapter-5: Number theory and Advance Encryption Standard

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2. Modular Arithmetic
3. Finite Fields of the Form $GF(p)$
4. Polynomial Arithmetic
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6. key expansion

Euclidian Algorithm

A method for finding the greatest common divisor (GCD) of two integers.

Step 1: Let a and b be two integers where $a > b$.

Step 2: Divide a by b and get the remainder r .

Step 3: Replace a with b and b with r .

Step 4: Repeat until $r=0$.

Step 5: The last non-zero remainder is the GCD.

Euclidian Algorithm

Finding the GCD of 56 and 15:

$$56 = 15 \times 3 + 11$$

$$15 = 11 \times 1 + 4$$

$$11 = 4 \times 2 + 3$$

$$4 = 3 \times 1 + 1$$

$$3 = 1 \times 3 + 0$$

GCD is 1.

Modular Arithmetic

- A system of arithmetic for integers, where numbers wrap around after reaching a certain value, the modulus.
- **Basic Operations**
 - Addition: $(a+b) \bmod n$
 - Subtraction: $(a-b) \bmod n$
 - Multiplication: $(a \times b) \bmod n$
 - Exponentiation: $(a^b) \bmod n$

Modular Arithmetic

- **Properties**
 - Congruences
 - Inverses
 - Residue Classes
-
- **Example**
 - $7 \bmod 5 = 2$
 - $18 \bmod 7 = 4$

Modular Arithmetic

Congruence

★ In cryptography, congruence ($\equiv$) instead of equality ($=$).

Examples:

$$15 \equiv 3 \pmod{12}$$

$$23 \equiv 11 \pmod{12}$$

$$33 \equiv 3 \pmod{10}$$

$$10 \equiv -2 \pmod{12}$$

$$\therefore a \equiv b \pmod{m}$$

$$\text{i.e. } a = km + b$$

	1	1	3
12	$\begin{array}{ c } \hline 15 \\ 12 \\ \hline 3 \end{array}$	$\begin{array}{ c } \hline 23 \\ 12 \\ \hline 11 \end{array}$	$\begin{array}{ c } \hline 33 \\ 30 \\ \hline 3 \end{array}$
	0	k	
12	$\begin{array}{ c } \hline 10 \\ 0 \\ \hline 10 \\ (-2) \end{array}$	$\begin{array}{ c } \hline a \\ \hline b \end{array}$	
		m	

Modular Arithmetic

Properties of Modular Arithmetic

Property	Expression
Commutative Laws	$(a + b) \bmod n = (b + a) \bmod n$ $(a \times b) \bmod n = (b \times a) \bmod n$
Associative Laws	$[(a + b) + c] \bmod n = [a + (b + c)] \bmod n$ $[(a \times b) \times c] \bmod n = [a \times (b \times c)] \bmod n$
Distributive Laws	$[a \times (b + c)] \bmod n = [(a \times b) + (a \times c)] \bmod n$
Identities	$(0 + a) \bmod n = a \bmod n$ $(1 \times a) \bmod n = a \bmod n$
Additive Inverse	For each $a \in \mathbb{Z}_n$, there exists a '-a' such that $a + (-a) \equiv 0 \bmod n$

Modular Arithmetic

Residue Classes

A residue class $[a]$ or $[a]_n$ is the set of integers congruent modulo n . (in following example $n=5$)

$$[0] = \{ \dots, -15, -10, -5, 0, 5, 10, 15, \dots \}$$

$$[1] = \{ \dots, -14, -9, -4, 1, 6, 11, 16, \dots \}$$

$$[2] = \{ \dots, -13, -8, -3, 2, 7, 12, 17, \dots \}$$

$$[3] = \{ \dots, -12, -7, -2, 3, 8, 13, 18, \dots \}$$

$$[4] = \{ \dots, -11, -6, -1, 4, 9, 14, 19, \dots \}$$

Modular Arithmetic

- **Cryptography**
- *RSA Algorithm*: Based on the difficulty of factoring large integers and the properties of modular exponentiation.
- *Diffie-Hellman Key Exchange*: Relies on modular arithmetic for secure key exchange.
- **Computer Science**
- *Hash Functions*: Use modular arithmetic to map data of arbitrary size to fixed-size values.
- *Random Number Generation*: Algorithms often use modular arithmetic to ensure uniform distribution.
- **Number Theory**
- Fermat's Little Theorem: $a^{p-1} \equiv 1 \pmod{p}$ for any integer a and prime p .
- *Chinese Remainder Theorem*: Provides a unique solution to simultaneous congruences with coprime moduli.

Finite Fields of the Form GF(p)

Group

A group G denoted by $\{G, \cdot\}$, is a set under some operations $(\cdot)$ if it satisfies the CAIN properties.

- ❖ C - Closure
- ❖ A - Associative
- ❖ I - Identity
- ❖ N - iNverse.

Finite Fields of the Form GF(p)

Group and Abelian Group

Property			Explanation
Abelian Group	Group	Closure	$a, b \in G$, then $(a \bullet b) \in G$.
		Associative	$a \bullet (b \bullet c) = (a \bullet b) \bullet c$ for all $a, b, c \in G$.
		Identity element	$(a \bullet e) = (e \bullet a) = a$ for all $a, e \in G$.
		Inverse element	$(a \bullet a') = (a' \bullet a) = e$ for all $a, a' \in G$.
	Commutative		$(a \bullet b) = (b \bullet a)$ for all $a, b \in G$.

Finite Fields of the Form GF(p)

Example

Question: Is $(\mathbb{Z}, +)$ a group?

Solution:

$\mathbb{Z} = \{ \dots, -3, -2, -1, 0, 1, 2, 3, \dots \}$ is an abelian group.

CAIN Property	Explanation	Satisfied?
Closure	If $a, b \in G$, then $(a \bullet b) \in G$. If $a = 5, b = -2 \in \mathbb{Z}$ then $(a + b) = -3 \in \mathbb{Z}$	✓
Associative	$a \bullet (b \bullet c) = (a \bullet b) \bullet c$ for all $a, b, c \in G$. $5 + (3 + 7) = (5 + 3) + 7 \in \mathbb{Z}$	✓
Identity element	$(a \bullet e) = (e \bullet a) = a$ for all $a \in G$. $(5 + 0) = (0 + 5) = 5$ for all $a \in G$.	✓
Inverse element	$(a \bullet a') = (a' \bullet a) = e$ for all $a, a' \in G$. $(5 + -5) = (-5 + 5) = 0$ for all $5, -5 \in \mathbb{Z}$	✓
Commutative	$(a \bullet b) = (b \bullet a)$ for all $a, b \in G$. $(5 + 9) = (9 + 5)$ for all $9, 5 \in \mathbb{Z}$.	✓

Finite Fields of the Form $GF(p)$

Notations

$\mathbb{N} \rightarrow$ Set of all natural numbers.

$\mathbb{W} \rightarrow$ Set of all whole numbers.

$\mathbb{Z} \rightarrow$ Set of all integers.

$\mathbb{C} \rightarrow$ Set of all complex numbers.

$\mathbb{Q} \rightarrow$ Set of all rational numbers.

$\mathbb{R} \rightarrow$ Set of all real numbers.

$\mathbb{Z}^+ \rightarrow$ Set of all positive integers.

$\mathbb{Z}^- \rightarrow$ Set of all negative integers.

Euclidian Algorithm

A finite field, also called Galois Field, is a field with a finite number of elements.

Characteristics

$GF(p)$ where p is a prime number

Elements: $\{0, 1, 2, \dots, p-1\}$

Addition and Multiplication modulo p .

Examples

$GF(2)$: $\{0, 1\}$

$GF(5)$: $\{0, 1, 2, 3, 4\}$

Finite Fields of the Form $\text{GF}(p)$

A1 - Closure	Group	Abelian Group	Ring	Commutative Ring	Integral Domain	Field
A2 - Associative						
A3 - Identity element						
A4 - Inverse element						
A5 - Commutativity of Addition						
M1 - Closure under multiplication						
M2 - Associativity of multiplication						
M3 - Distributive						
M4 - Commutativity of multiplication						
M5 - Multiplicative Identity						
M6 - No Zero Divisors						
M7 - Multiplicative Inverse						

Polynomial Arithmetic

- Operations on polynomials including addition, subtraction, multiplication, and division.
- Key Concepts
- Addition/Subtraction: Combine like terms.
- Multiplication: Distribute terms.
- Division: Divide leading terms, multiply and subtract until remainder is of lower degree.
- Degree: Highest power of the variable in the polynomial

Polynomial Arithmetic in Finite Fields

- Performing polynomial operations in $GF(p)$.
- Irreducible Polynomials : Polynomials that cannot be factored into polynomials of lower degree over $GF(p)$.

Addition

$\oplus$	0	1
0	0	1
1	1	0

Multiplication

$\otimes$	0	1
0	0	0
1	0	1

Additive Inv

	a	0	1
-a	0	0	1

Multiplicative Inv

	a	0	1
$1/a$	-	-	1

$$(ii) GF(5) = \{0, 1, 2, 3, 4\}$$

$$a + (-a) = 0$$

$$a + (\frac{1}{a}) = 1$$

Addition

$\oplus$	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	$5 \bmod 5 = 0$
2	2	3	4	$5 \bmod 5 = 0$	$6 \bmod 5 = 1$
3	3	4	$5 \bmod 5 = 0$	$6 \bmod 5 = 1$	$7 \bmod 5 = 2$
4	4	$5 \bmod 5 = 0$	$6 \bmod 5 = 1$	$7 \bmod 5 = 2$	$8 \bmod 5 = 3$

Multiplication

$\otimes$	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	$6 \bmod 5 = 1$	$8 \bmod 5 = 3$
3	0	3	$6 \bmod 5 = 1$	$9 \bmod 5 = 4$	$12 \bmod 5 = 2$
4	0	4	$8 \bmod 5 = 3$	$12 \bmod 5 = 2$	$16 \bmod 5 = 1$

Additive Inv

	a	0	1	2	3	4
-a	0	4	3	2	1	0

Multiplicative Inv

	a	0	1	2	3	4
-a	-	-	1	3	2	4

Polynomial Arithmetic - Usages

Error-Correcting Codes

Reed-Solomon Codes: Used in data transmission and storage for error detection and correction.

Cryptography

Advanced Encryption Standard (AES): Uses finite fields for encryption and decryption.

Algebraic Geometry

Elliptic Curves: Used in cryptographic algorithms like ECC (Elliptic Curve Cryptography).

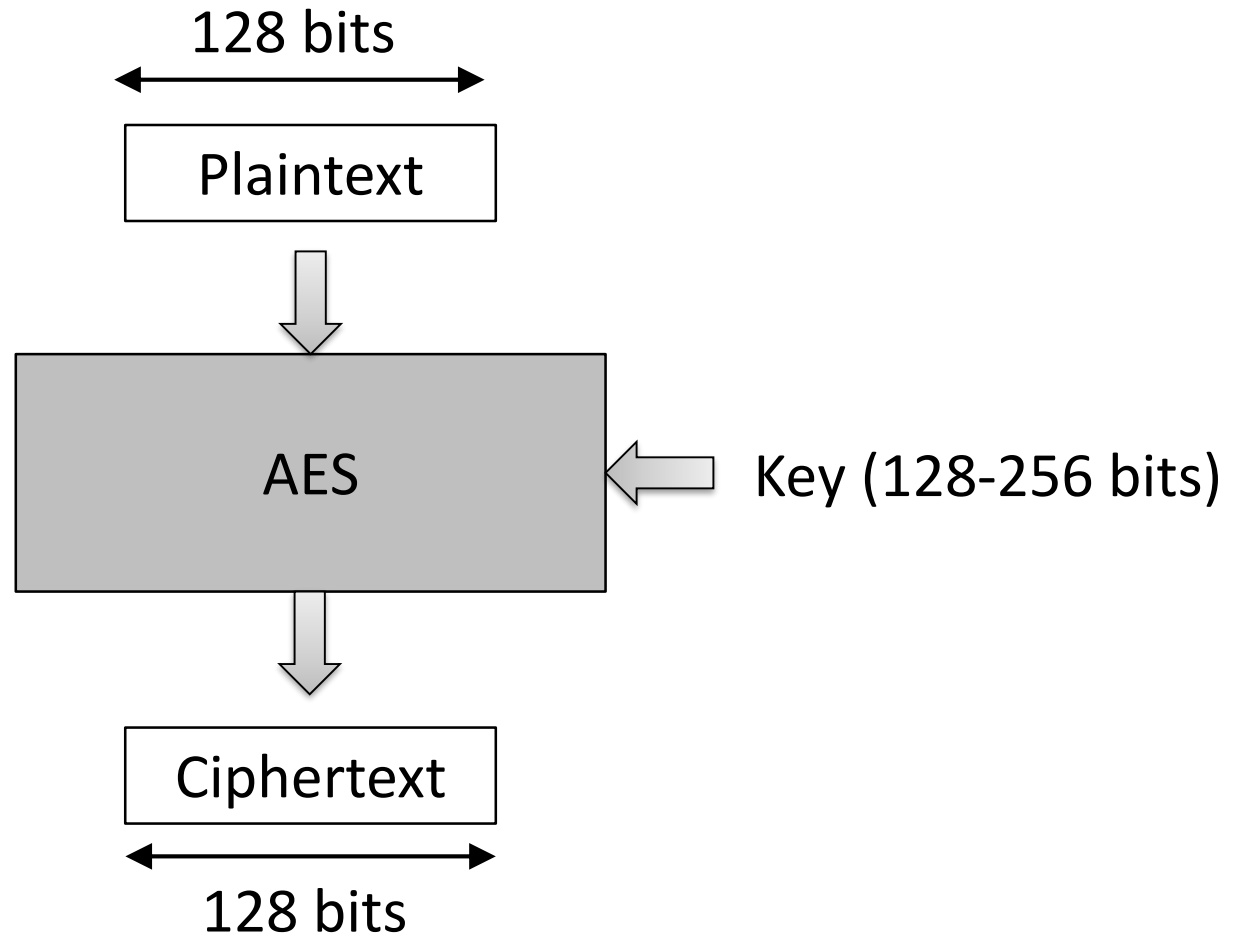
Advance Encryption Standard

- The Rijndael proposal for AES defined a cipher in which the block length and the key length can be independently specified to be 128, 192, or 256 bits.

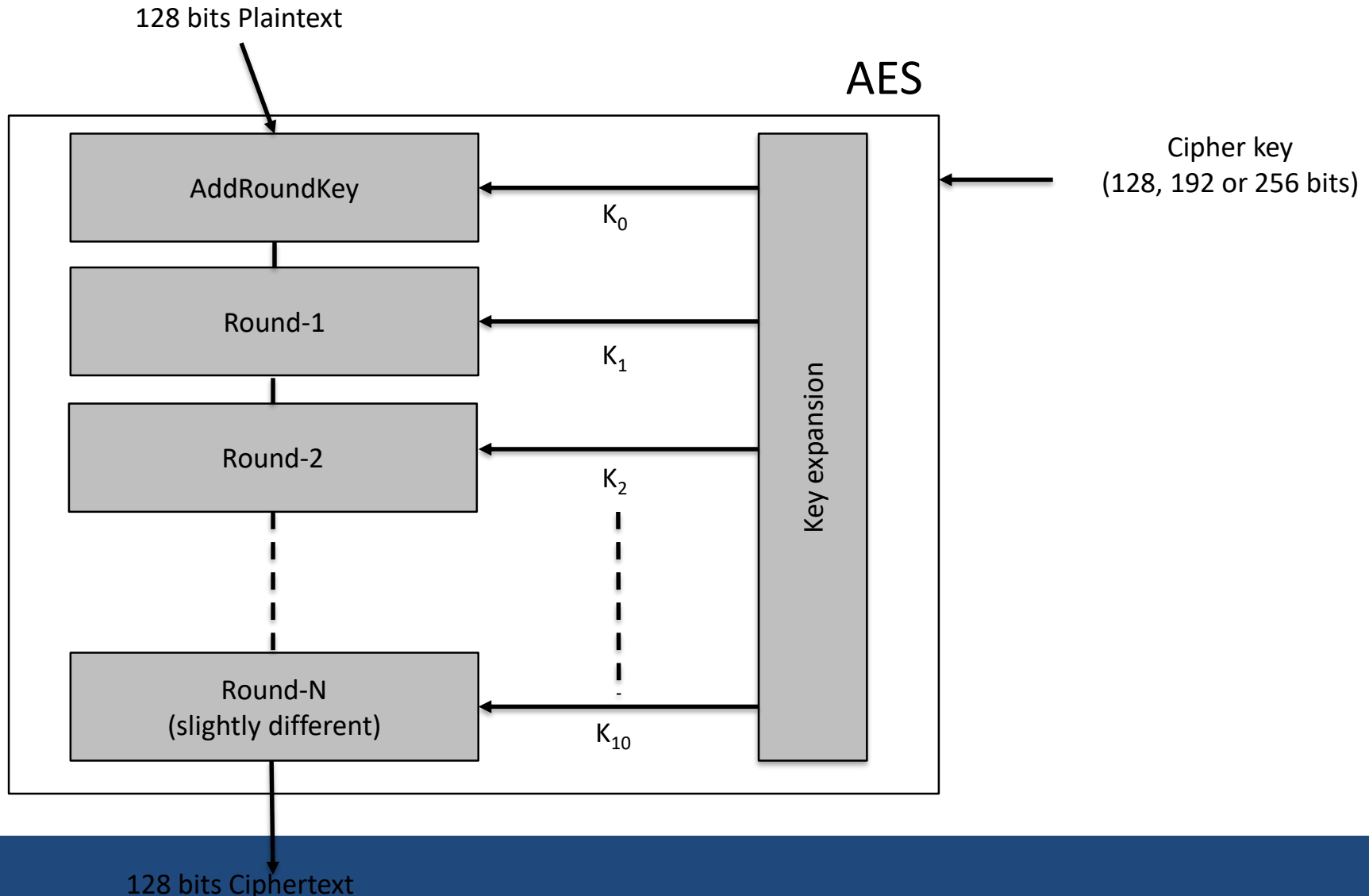
Key size (words/ bytes/ bits)	4/16/128	6/24/192	8/32/256
Block size (words/ bytes/ bits)	4/16/128	4/16/128	4/16/128
Round key size (words/ bytes/ bits)	4/16/128	4/16/128	4/16/128
Number of Rounds	10	12	14

- AES designed to have characteristics
- Resistance against all known attacks
- Speed and code compactness on a wide range of platforms
- Design simplicity

Advance Encryption Standard



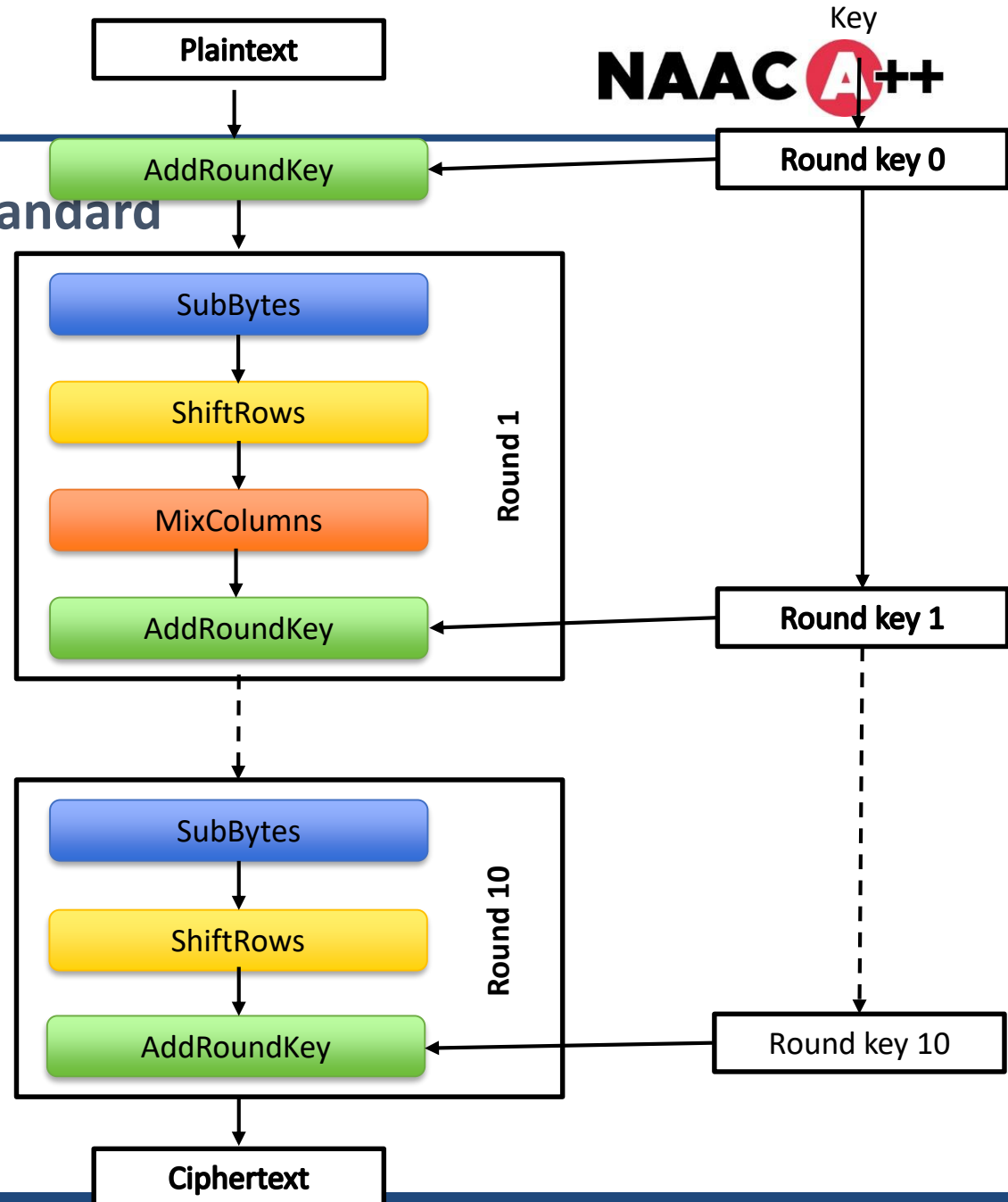
Advance Encryption Standard



Advance Encryption Standard

Notes:

1. One AddRoundKey is applied before the First round.
2. The third transformation is missing in the last round



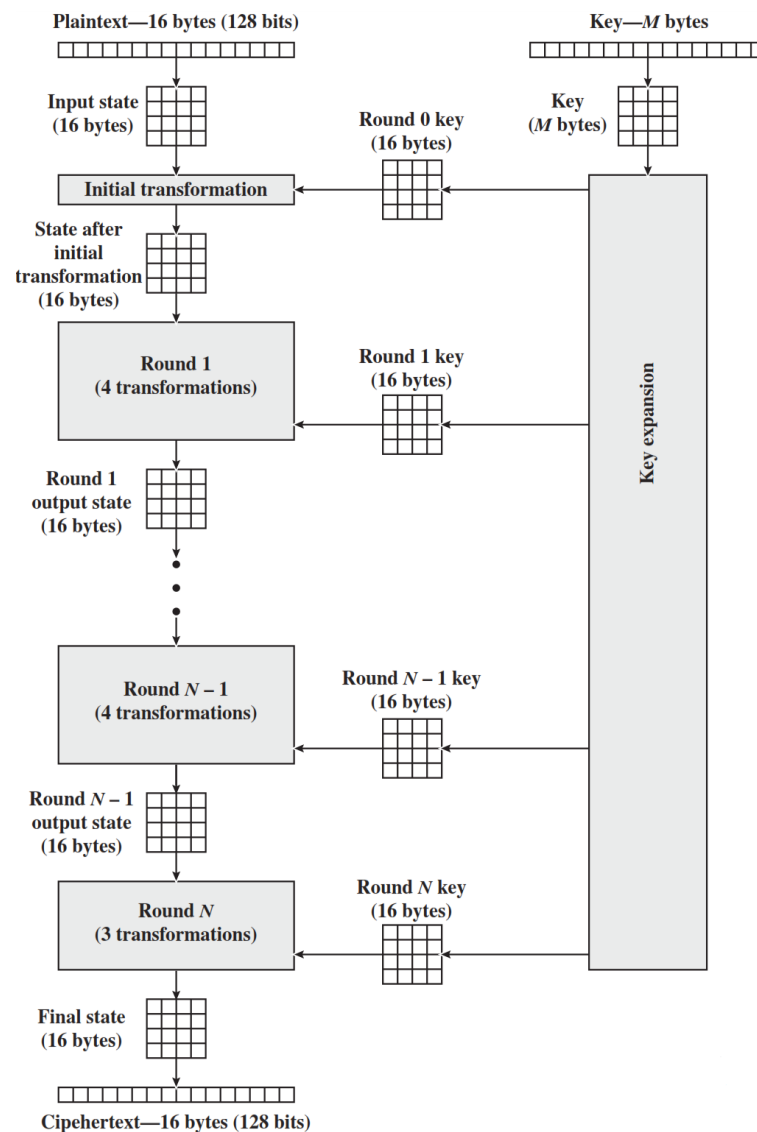
Advance Encryption Standard

Initialization

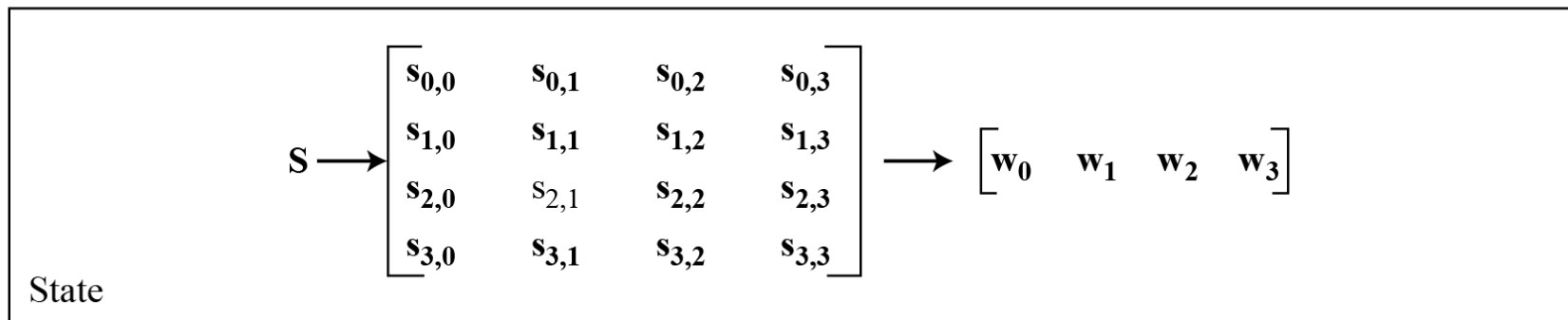
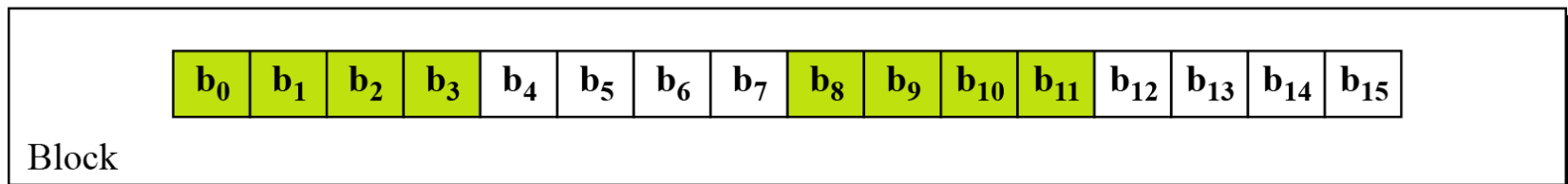
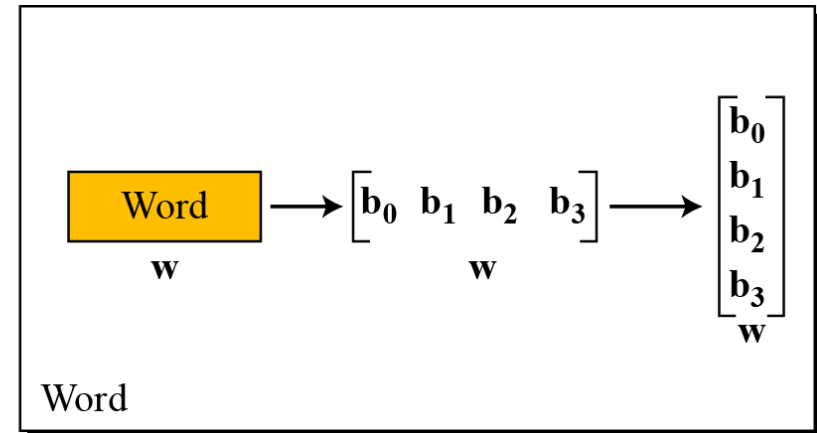
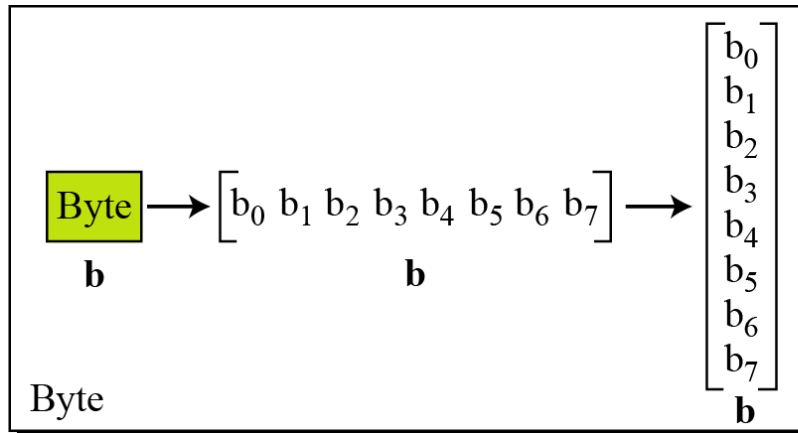
- Expand 16-byte key to get the actual key block to be used.
- Initialize 16-byte plaintext block called as state.
- XOR the state with the key block.

For each round

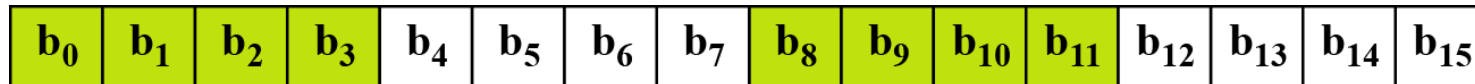
- Apply S-box
- Rotate rows of state
- Mix columns
- Add Round key: XOR the state with key block.



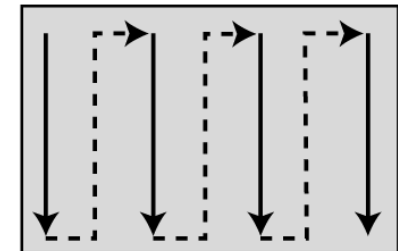
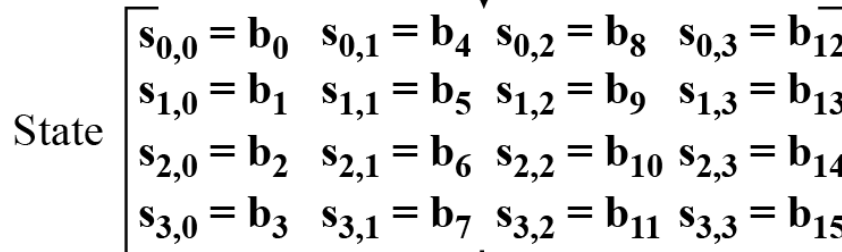
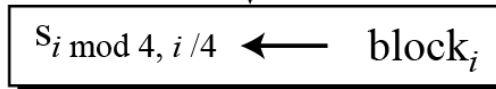
Advance Encryption Standard



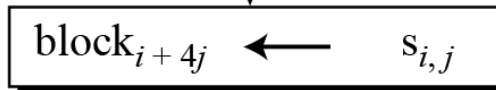
Advance Encryption Standard



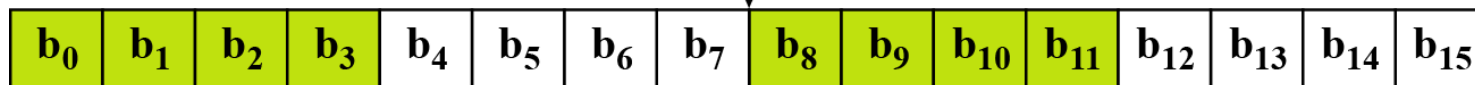
Block



Insertion and extraction flow



Block



Advance Encryption Standard

Text

A E S U S E S A M A T R I X **Z Z**

Hexadecimal

00 04 12 14 12 04 12 00 0C 00 13 11 08 23 19 19

00	12	0C	08	State
04	04	00	23	
12	12	13	19	
14	00	11	19	

Advance Encryption Standard

- The first N-1 rounds consist of four distinct transformation functions.

SubBytes

- The 16 input bytes are substituted using an S-box

ShiftRows

- Each of the four rows of the matrix is shifted to the left

MixColumns

- Each column of four bytes is now transformed using a special mathematical function.

AddRoundKey

- The 16 bytes of the matrix are now considered as 128 bits and are XORed to the 128 bits of the round key.

Advance Encryption Standard

State:

32	88	31	e0
43	5a	31	37
f6	30	98	07
a8	8d	a2	34

Cipher key:

2b	28	ab	09
7e	ae	f7	cf
15	d2	15	4f
16	a6	88	3c

Advance Encryption Standard

AddRoundKey: input state $\oplus$ Cipher key

32	88	31	e0
43	5a	31	37
f6	30	98	07
a8	8d	a2	34

 $\oplus$

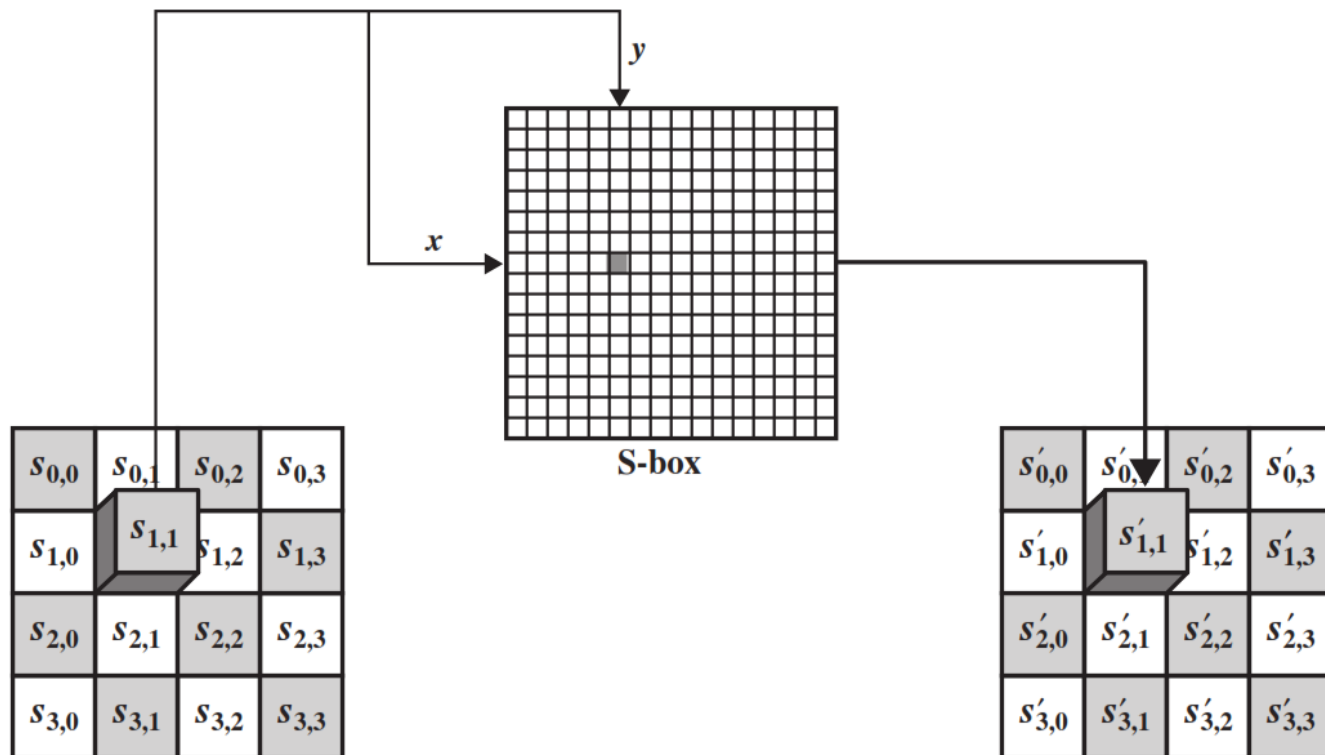
2b	28	ab	09
7e	ae	f7	cf
15	d2	15	4f
16	a6	88	3c

 $=$

19	a0	9a	e9
3d	f4	c6	f8
e3	e2	8d	48
be	2b	2a	08

Advance Encryption Standard

- The forward substitute byte transformation, called SubBytes, is a simple table lookup



Advance Encryption Standard

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
B	E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

Input: 19

Output: D4

Advance Encryption Standard

Input for SubByte

19	a0	9a	e9
3d	f4	c6	f8
e3	e2	8d	48
be	2b	2a	08

Output of SubByte

d4	e0	b8	le
27	bf	b4	41
11	98	5d	52
ae	f1	e5	30

Advance Encryption Standard

- The first row of State is not altered.
- For the second row, a 1-byte circular left shift is performed.
- For the third row, a 2-byte circular left shift is performed.
- For the fourth row, a 3-byte circular left shift is performed.

d4	e0	b8	le	← No rotation
27	bf	b4	41	← Rotate 1 byte
11	98	5d	52	← Rotate 2 bytes
ae	f1	e5	30	← Rotate 3 bytes

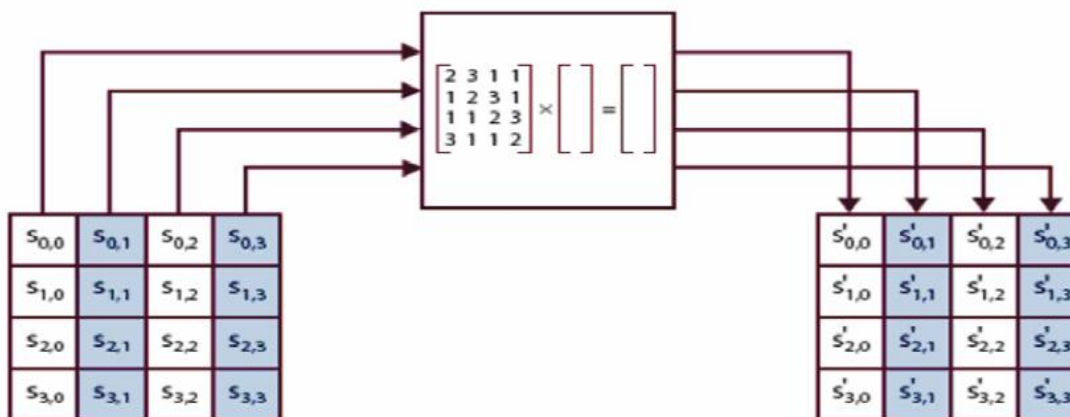
Input for ShiftRows

d4	e0	b8	le
bf	b4	41	27
5d	52	11	98
30	ae	f1	e5

Output of ShiftRows

Advance Encryption Standard

- Each byte of a column is mapped into a new value that is a function of all four bytes in that column.
- Constant matrices used by MixColumns.



$$\begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} \begin{bmatrix} s_{0,0} & s_{0,1} & s_{0,2} & s_{0,3} \\ s_{1,0} & s_{1,1} & s_{1,2} & s_{1,3} \\ s_{2,0} & s_{2,1} & s_{2,2} & s_{2,3} \\ s_{3,0} & s_{3,1} & s_{3,2} & s_{3,3} \end{bmatrix} = \begin{bmatrix} s'_{0,0} & s'_{0,1} & s'_{0,2} & s'_{0,3} \\ s'_{1,0} & s'_{1,1} & s'_{1,2} & s'_{1,3} \\ s'_{2,0} & s'_{2,1} & s'_{2,2} & s'_{2,3} \\ s'_{3,0} & s'_{3,1} & s'_{3,2} & s'_{3,3} \end{bmatrix}$$

Advance Encryption Standard

d4	e0	b8	le
bf	b4	41	27
5d	52	11	98
30	ae	f1	e5

 $\bullet$

$$\begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} =$$

04	e0	48	28
66	cb	f8	06
81	19	d3	26
e5	9a	7a	4c

$$\begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} \bullet \begin{bmatrix} d4 \\ bf \\ 5d \\ 30 \end{bmatrix} = \begin{bmatrix} 04 \\ 66 \\ 81 \\ e5 \end{bmatrix}$$

Advance Encryption Standard

- In the forward add round key transformation, the 128 bits of State are bitwise XORed with the 128 bits of the round key.

04	e0	48	28
66	cb	f8	06
81	19	d3	26
e5	9a	7a	4c

 $\oplus$

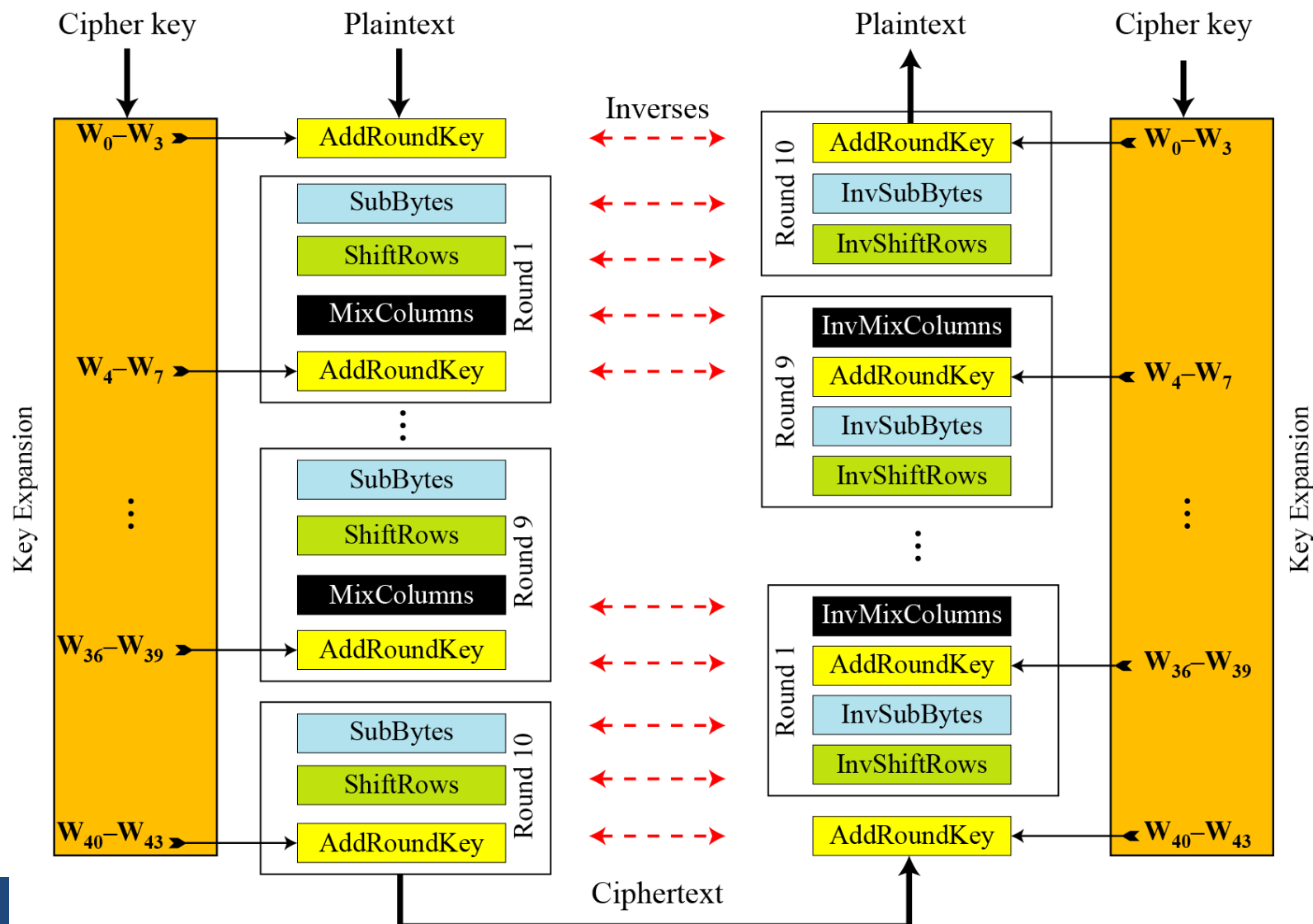
a0	88	23	2a
fa	54	a3	6c
fe	2c	39	76
17	b1	39	05

 $=$

A4	68	6b	02
9c	9f	5b	6a
7f	35	Ea	50
F2	2b	43	49

04	a0	A4
66	fa	9c
81	fe	7f
e5	17	F2

Advance Encryption Standard



AES – Key Expansion

Words for each round

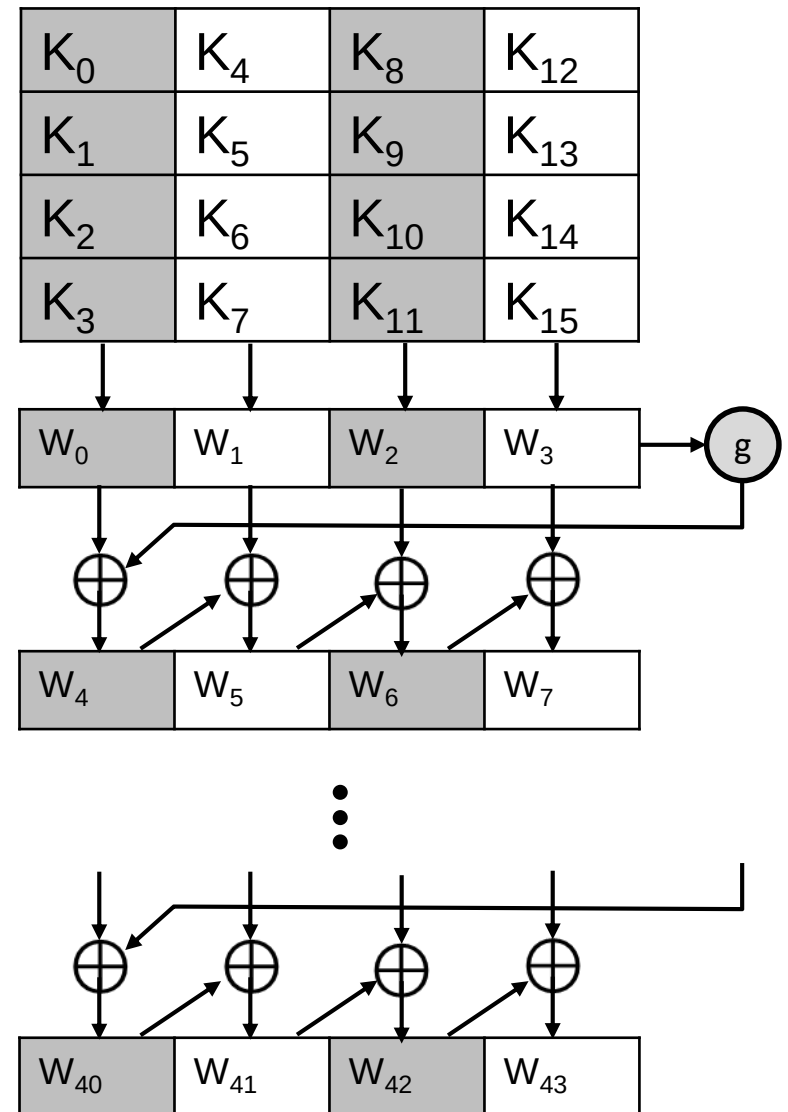
Round	Words			
Pre-round	W_0	W_1	W_2	W_3
Round 1	W_4	W_5	W_6	W_7
Round 2	W_8	W_9	W_{10}	W_{11}
...	...			
Round N	W_{40}	W_{41}	W_{42}	W_{43}

AES – Key Expansion

The AES key expansion algorithm takes as input a four-word (16-byte) key and produces a linear array of **44 words (176 bytes)**.

Each added word $w[i]$ depends on the immediately preceding word, $w[i - 1]$.

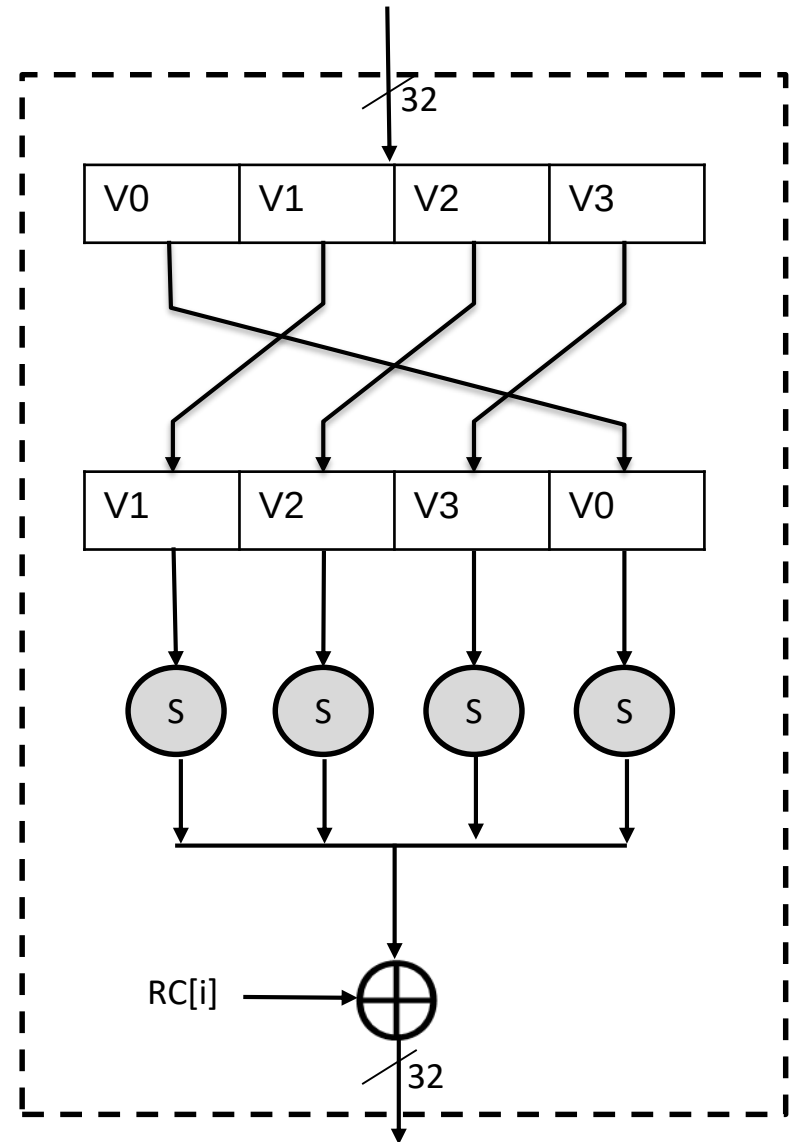
In three out of four cases, a simple XOR is used.



AES – Key Expansion

Rcon Table

1	2	3	4	5	6	7	8	9	10
01	02	04	08	10	20	40	80	1b	36
00	00	00	00	00	00	00	00	00	00
00	00	00	00	00	00	00	00	00	00
00	00	00	00	00	00	00	00	00	00



AES – Key Expansion

Plaintext:	0123456789abcdef fedcba9876543210
Key:	0f1571c947d9e8590cb7add6af7f6798
Ciphertext:	ff0b844a0853bf7c6934ab4364148fb9

Key Words	Auxiliary Function
$w_0 = 0f\ 15\ 71\ c9$ $w_1 = 47\ d9\ e8\ 59$ $w_2 = 0c\ b7\ ad\ d6$ $w_3 = af\ 7f\ 67\ 98$	$RotWord(w_3) = 7f\ 67\ 98\ af = x_1$ $SubWord(x_1) = d2\ 85\ 46\ 79 = y_1$ $Rcon(1) = 01\ 00\ 00\ 00$ $y_1 \oplus Rcon(1) = d3\ 85\ 46\ 79 = z_1$
$w_4 = w_0 \oplus z_1 = dc\ 90\ 37\ b0$ $w_5 = w_4 \oplus w_1 = 9b\ 49\ df\ e9$ $w_6 = w_5 \oplus w_2 = 97\ fe\ 72\ 3f$ $w_7 = w_6 \oplus w_3 = 38\ 81\ 15\ a7$	$RotWord(w_7) = 81\ 15\ a7\ 38 = x_2$ $SubWord(x_2) = 0c\ 59\ 5c\ 07 = y_2$ $Rcon(2) = 02\ 00\ 00\ 00$ $y_2 \oplus Rcon(2) = 0e\ 59\ 5c\ 07 = z_2$
$w_8 = w_4 \oplus z_2 = d2\ c9\ 6b\ b7$ $w_9 = w_8 \oplus w_5 = 49\ 80\ b4\ 5e$ $w_{10} = w_9 \oplus w_6 = de\ 7e\ c6\ 61$ $w_{11} = w_{10} \oplus w_7 = e6\ ff\ d3\ c6$	$RotWord(w_{11}) = ff\ d3\ c6\ e6 = x_3$ $SubWord(x_3) = 16\ 66\ b4\ 83 = y_3$ $Rcon(3) = 04\ 00\ 00\ 00$ $y_3 \oplus Rcon(3) = 12\ 66\ b4\ 8e = z_3$
$w_{12} = w_8 \oplus z_3 = c0\ af\ df\ 39$ $w_{13} = w_{12} \oplus w_9 = 89\ 2f\ 6b\ 67$ $w_{14} = w_{13} \oplus w_{10} = 57\ 51\ ad\ 06$ $w_{15} = w_{14} \oplus w_{11} = b1\ ae\ 7e\ c0$	$RotWord(w_{15}) = ae\ 7e\ c0\ b1 = x_4$ $SubWord(x_4) = e4\ f3\ ba\ c8 = y_4$ $Rcon(4) = 08\ 00\ 00\ 00$ $y_4 \oplus Rcon(4) = ec\ f3\ ba\ c8 = 4$

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